Muhammed Shebin V A

AI Engineer | Full-Stack Developer (Python, React) | Agentic AI & Generative AI
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PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with 4 years at Infosys, architecting production Agentic AI systems for enterprise IT operations and security. Currently building a governed, multi-tenant vulnerability-remediation platform (Python, FastAPI, PostgreSQL, Temporal) with a deterministic policy engine and a fail-closed AI gateway, alongside multi-agent ITSM analytics tools (LangGraph, LangChain, Azure OpenAI GPT-4o) spanning RAG, semantic search, and automated ticket-quality scoring. Builds full-stack React/TypeScript dashboards end-to-end and uses AI-assisted development tools (Claude Code, GitHub Copilot) to ship faster.

TECHNICAL SKILLS

- **AI & GenAI:** Agentic AI, Multi-Agent Systems, LangGraph (StateGraph & ReAct), LangChain, Azure OpenAI (GPT-4o), LLM Orchestration, Prompt Engineering, RAG, Vector Databases, OpenAI Embeddings, Semantic Search (SBERT, BM25, Cross-Encoder Reranking), Hybrid Search, MCP (Model Context Protocol)
- **AI Governance & Guardrails:** Deterministic Policy Engines, Human-in-the-Loop Approval Gating, Confidence-Based Automation Controls, Guardrails (PII/Secret Redaction, Prompt-Injection Defense, Topic Gating)
- **Backend & APIs:** Python, FastAPI, Flask, Node.js, RESTful APIs, JWT, RBAC, Pydantic, Microservices, Temporal (Workflow Orchestration)
- **Frontend:** React.js, TypeScript, JavaScript (ES6+), Redux Toolkit, Tailwind CSS, HTML5, CSS3
- **Databases & Caching:** PostgreSQL (incl. Row-Level Security), DuckDB (VSS), Redis, PL/SQL, SQL
- **DevOps & Tools:** Docker, Git, GitHub
- **Hosting & Deployment:** IIS (Windows Server), Apache httpd (Linux), NSSM (Windows Services), Virtual Machines
- **AI-Assisted Development:** Claude Code, GitHub Copilot
- **Integrations:** ServiceNow REST API, Jira, SolarWinds

PROFESSIONAL EXPERIENCE

Infosys Limited | Bengaluru, India **October 2022 – Present**
Senior Systems Associate
Tech Stack: *Python, FastAPI, Flask, LangGraph, LangChain, Azure OpenAI GPT-4o, MCP, React.js, TypeScript, PostgreSQL, DuckDB, ServiceNow API, Docker*

- **Current Initiative:** Architecting FVM (REMORA), a multi-tenant AI-assisted vulnerability remediation orchestration platform unifying vulnerability scanners, CMDB, EDR, and ticketing tools into a single governed workflow, replacing manual, spreadsheet-driven patch coordination.
- **Policy & Governance:** Designed a deterministic policy engine that evaluates every remediation action against risk, ownership, SLA, and blast-radius rules before execution, with a network-isolated, fail-closed Azure OpenAI gateway used only for read-only investigation and reporting.
- **Agentic AI:** Built stateful multi-agent workflows with LangGraph, LangChain, and Azure OpenAI GPT-4o for context-aware classification, risk analysis, and intelligent routing of unstructured ITSM ticket data.
- **AI Platform Delivery:** Played a lead role in building Cognitive Analyzer, an AI ticket-analytics platform now deployed across 20+ global enterprise accounts, delivering measured incident-ticket reductions of 11–49% and service-request reductions of 8–62% (internal deployment metrics, July 2026).
- **Performance Optimization:** Currently improving Cognitive Analyzer's ticket-categorization pipeline, optimizing SBERT inference with ONNX Runtime and caching embeddings in DuckDB to speed up large-volume ticket batch processing.
- **RAG & Semantic Search:** Developed RAG and semantic-search pipelines using embeddings and vector databases (DuckDB VSS), powering an on-premise, RAG-grounded AI chatbot for natural-language analytics.
- **Backend & APIs:** Engineered secure Python (FastAPI/Flask) REST APIs with JWT/RBAC and MCP integrations connecting ServiceNow, Jira, and SolarWinds.
- **Full-Stack Development:** Built React (TypeScript, Redux Toolkit, Tailwind CSS) single-page dashboards with interactive analytics, reducing executive reporting turnaround from days to seconds.
- **DevOps:** Containerized applications with Docker for consistent, repeatable deployments across environments.

- **Hosting & Deployment:** Host and maintain web applications and APIs on non-production virtual machines, using IIS on Windows and Apache httpd on Linux, and run backend services continuously as Windows services with NSSM.
- **AI-Assisted Development:** Use Claude Code and GitHub Copilot to accelerate code generation, refactoring, debugging, and test writing across Python and React codebases.

KEY PROJECTS

FVM (REMORA) – AI-Assisted Vulnerability Remediation Orchestration Platform (Sole Developer, Current)

Tech: Python, FastAPI, PostgreSQL (RLS), Temporal, Redis, Docker, React, TypeScript, Azure OpenAI

- Architecting a multi-tenant platform that unifies vulnerability scanners, CMDB, EDR, and ticketing tools into one governed remediation pipeline (triage, ownership, approval, execution, verification, evidence), replacing manual, spreadsheet-driven patch coordination.
- Designed a deterministic policy engine enforcing risk, ownership, SLA, and blast-radius rules before any remediation action executes; AI is used only for read-only investigation and summarization, with no network path to trigger or approve actions.
- Built a measured-autonomy model that grants automation trust per asset class based on real success and rollback history rather than manual configuration.
- Built a configuration-driven, multi-tenant platform (SLA policy, approval routing, maintenance windows, connector registry, tenant onboarding), using Row-Level Security in PostgreSQL for tenant isolation and Redis for locking and rate-limiting.
- Designed tamper-evident evidence records tied to every remediation action for audit and compliance reporting.

Cognitive Analyzer – Intelligent Ticket Analytics Platform (Lead Contributor)

Tech: Python, Flask, LangGraph, LangChain, DuckDB, React, TypeScript, Docker

- Three-layer categorization pipeline (keyword matching, SBERT semantic similarity, and LLM-assisted residual clustering) ingesting Excel and ServiceNow data to surface ticket trends at scale.
- Composite service-health scoring (SLA, MTTR, outlier rate, backlog trend) producing a Red/Amber/Green verdict and quadrant analysis to flag quick wins vs. critical issues.
- Built a knowledge-base accuracy pipeline combining BM25 keyword search, SBERT semantic similarity, and a cross-encoder reranker to score how well documentation matches real tickets.
- Interactive React dashboard with a LangGraph agent for natural-language Q&A, enabling self-service ticket analytics for support teams.

CLARA – Compliance & Log Assessment Review Agent (AI ITSM QA Co-Pilot) (Contributor)

Tech: Python, FastAPI, LangGraph, Azure OpenAI GPT-4o, OpenAI Embeddings, DuckDB, MCP, React, TypeScript

- Built a rule-based scoring engine (regex and structure analysis, no AI cost) that rates every work note across five quality dimensions, then used GPT-4o to turn low scores into plain-English coaching narratives; the platform delivers 100% ticket-quality coverage with audits completed in seconds and up to 80% reduction in manual audit effort (company-wide deployment results).
- Built a hybrid similarity engine (structural match, embedding-based text similarity, and context overlap) to detect duplicate and related tickets and surface problem-record candidates.
- Delivered triage queues, AI-generated coaching narratives, and executive summaries for real-time visibility into ticket quality.

SAGA – Service Assistance & Governance Agent (Minor Contributor)

- Contributed to SAGA, a multi-agent ITSM co-pilot exploring automated ticket triage, SLA-breach prediction, and auto-resolution through a React dashboard.

EDUCATION

Master of Computer Applications (MCA) – Online **Jan 2024 – Mar 2026**
University of Mysore, Karnataka, India

Bachelor of Science in Computer Science **June 2019 – July 2022**
Bharathiar University, Tamil Nadu, India

CERTIFICATIONS

- Infosys Certified Generative AI Professional – Intermediate
- Infosys Certified Expert – Natural Language Processing Professional
- Infosys Topaz Fabric – 101 (TOC / Lex)
- AZ-900: Microsoft Azure Fundamentals